

Experimental study on the dynamic response of a 3-D wedge under asymmetric impact

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Abstract: Water entry problems represent complex multiphase flows involving air, water, and structure interaction, occurring rapidly in rough seas, and potentially effecting structural integrity of floating structures. This paper experimentally investigates asymmetric slamming loads acting on a 3-D elastic wedge section. The specimen, featuring two different bottom plates (stiffened and unstiffened), each 4 mm thick, aims to assess the effect of structural stiffness on dynamic loads. The experiments are conducted at different drop heights of 25 cm and 50 cm and varying heel angles from 5° to 25°. The paper describes the experimental conditions, including wedge geometry, material properties, and the test plan. The study explores the influence of heel angle on impact acceleration, revealing an increase in peak acceleration with a higher inclination angle, particularly in the vertical direction. Additionally, the hydrodynamic pressure resulting from asymmetric slamming is presented. The pressure results analyzed and compared at different locations along the length of the wedge. The experimental findings indicate that, despite the leeward side (stiffened) experiencing a smaller hydrodynamic load, the heel angle significantly affects pressure results on the windward side (unstiffened), leading to a more pronounced dynamic response. The time history of pressure results emphasizes the effect of elastic vibrations, particularly noticeable on the unstiffened bottom plate. This study contributes to a deeper understanding of asymmetric slamming on aluminum structures, facilitating the enhancement of mathematical models and the validation of numerical simulations.

Key words: Flexible fluid structure interactions (FFSI), experimental study, asymmetric impact, hydrodynamic pressure, slamming loads

0. Introduction

Ships and offshore structures that operate in severe sea conditions encounter many challenges concerning their structural integrity and the surrounding environment. A critical phenomenon demanding comprehensive understanding is slamming, which occurs when a ship or a part of structure forcefully interacts with oncoming waves during extreme sea conditions, posing a significant risk of causing structural damage.

Bow slamming is a specific form of ship slamming that occurs at the bow of a vessel when it encounters oncoming waves in rough seas. This phenomenon is particularly common in moderate to high-speed crafts or vessels designed for offshore operations, where the bow is more likely to impact waves directly. In this situation, the water close to the boundary experiences a sudden acceleration, leading

to increased pressure and significant forces. The magnitude of the pressure impact is closely linked to the angle between the structure and the water surface, with a notable increase at smaller relative angles. To assess the impact loads on the bow section entering the water surface, it becomes imperative to determine the hydrodynamic pressure distribution around the structure. This necessitates a comprehensive analysis of the fluid dynamics to understand how the water pressure interacts with the solid body during the slamming event.

The initial study on water entry issues was introduced by von Karman^[1], Wagner^[2], who formulated a mathematical model using ideal flow assumptions to account for the effect of vertical velocity on the flow field and resulting hydrodynamic pressure. Subsequently, multiple studies have proposed various mathematical models based on the Wagner theory^[3-7]. Tassin et al.^[8] conducted an extensive investigation on several analytical models for water entry problems of rigid bodies and compared the findings with an asymptotic method presented by Zhao and Faltinsen^[9]. Since the slamming phenomenon involves complex

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interactions between air, water, and structure, simplifying the problem as a 2-D rigid body impact may not always provide accurate insights. The assumption of a rigid body implies that the hydrodynamic pressure remains unaffected by structural responses. Consequently, the complicated interaction known as hydroelasticity, involving the interplay between hydrodynamic loads and elastic deformations, cannot be adequately considered under such assumptions. Khabakhpasheva and Korobkin^[10] presented a semi-analytical method for the analysis of the water entry problem of a two-dimensional elastic wedge. The modeling of the wedge structure was carried out based on Euler beams, and the fluid flow was assumed to be two-dimensional, symmetric, and potential. Within their model, the hydrodynamic loads were calculated using Wagner water impact theory. Shams and Porfiri^[11] conducted an analysis of the hydroelastic impact of flexible wedges entering the water surface under conditions of both free fall and constant speed. The flow characteristics were described using the Wagner theory, and the wedge was modelled employing the Euler-Bernoulli beam theory. While their proposed model effectively predicted the influence of structural flexibility on hydrodynamic loads, it was observed that the model tended to overestimate wedge deformation and the total impact force. The hydroelastic slamming of a 2-D flexible body was modelled by Sun et al.^[12] utilizing the modified Logvinovich model (MLM), which accounts for the elastic deflections of the body surface to calculate hydrodynamic loads. The model's accuracy was validated by comparing results with previous experimental data. While there was good agreement between the results obtained with the developed model and previous data, it was observed that the presented semi-analytical model cannot manage the large flow deformation during the deep penetration stage.

In recent decades, alongside analytical models^[13-15], numerous numerical methods have been developed to provide a more precise simulation of the complex physics involved in the water entry problem for elastic structures^[16-21]. These numerical approaches aim to comprehensively address the fully coupled nonlinear phenomenon associated with elastic water entry. The advancement of computational techniques has played a pivotal role in capturing the complexities of flexible fluid-structure interactions during water entry events. These numerical methods, such as finite element method (FEM) and computational fluid dynamics (CFD), offer a valuable tool to explore the dynamic aspects of water entry problems, allowing for a more in-depth understanding of the behavior of elastic structures subjected to slamming conditions. Maki et al.^[22] employed a combination of CFD and FEM techniques to predict the hydroelastic response of a

wedge structure. Utilizing a one-way coupling method, they compared the maximum displacement of the elastic wedge during both the impact and penetration stages with previously published results. Wang and Guedes Soares^[23] discovered the significance of hydroelasticity in influencing local slamming-induced responses on a rectangular plate by examining the relationship between the loading period and the natural vibration period of the structure. A multi material arbitrary Eulerian Lagrangian (MMALE) method was employed by Truong et al.^[24] to calculate slamming loads on a flat stiffened plate and analyze its dynamic response. Later, Truong et al.^[25] conducted a benchmark study to evaluate the accuracy of various fluid-structure interaction (FSI) coupling techniques in predicting slamming loads and responses for stiffened flat plates. The assessment involved four distinct numerical methods, comparing the impact loads with the corresponding bottom deflections at different vertical velocities. Hosseinzadeh and Tabri^[26] conducted a numerical simulation of the water entry problem for a 2-D elastic wedge using a two-way coupling technique and compared the outcomes for both free fall and constant impact velocity scenarios with experimental data. The further development of hydroelastic slamming was achieved by coupling CFD-FEM methods on flat flexible plates, enabling a detailed investigation into the effect of structural rigidity on peak pressure^[27]. Moreover, Yan et al.^[28] conducted a comprehensive evaluation of FSI simulations against experimental data, emphasizing hydroelasticity and air trapping impacts, numerical uncertainties, and the reliability of modeling assumptions in predicting bottom slamming. Recently Hosseinzadeh et al.^[29] presented an in-depth exploration of hydroelastic slamming in 3-D V-shaped structures, employing two different numerical models and comparing the results with experimental data. In the latter publication, the authors studied the challenges associated with each method, investigated the influence of flexural rigidity on hydrodynamic loads and structural response, emphasized the significance of fully coupled FSI simulations, and discussed the implications of three-dimensionality on the hydroelastic slamming.

Although the water entry problems have been extensively studied for both rigid and elastic structures over the decades, research on slamming events under oblique sea conditions has been comparatively limited. The difficulties involved in oblique water entry through elastic structures lead to a lack of studies in this particular area. The investigation of oblique slamming is crucial as it may result in asymmetric flow field patterns, posing unique challenges to ship structural integrity^[30-33]. Mesa and Maki^[34] introduced a FSI algorithm, specifically presented for a flat plate

entering water with high horizontal velocity. Later on, Izadi et al.^[35] conducted a numerical investigation into the hydroelastic impact of symmetric and asymmetric 2-D wedge sections. Their findings revealed that the heel angle exerts a significant influence on both hydrodynamic pressure and structural deflection. Xie et al.^[36] employed an uncoupled CFD-FEM approach to investigate the dynamic response of a truncated ship-hull structure subjected to asymmetrical slamming. They validated the ship motions, slamming loads, and dynamic response against experimental data. A parametric study on the water entry of a symmetrical and asymmetrical 2-D wedge into waves was presented by Wang et al.^[37], during which they reported that the influence of the horizontal velocity of the wave on the wedge is nearly negligible when the side with the smaller deadrise angle faces the incoming wave. The oblique slamming process of an elastic 2-D wedge was analyzed by Tavakoli and Hirdaris^[38], and it was found that the resulting dynamic response and equivalent stresses were more noticeable.

The experimental study plays a crucial role in obtaining a deeper understanding of slamming-induced impacts and validating analytical and numerical models. Past experimental investigations into water entry, involving both rigid and elastic bodies, have utilized various structures such as flat plates, prismatic wedges, cylinders, and even models resembling ship hulls or truncated ship hulls^[39-41]. Shams et al.^[42] conducted an experimental study characterizing the water entry of an asymmetric wedge into a quiescent fluid. Their findings revealed that as the heel angle increases, the location of the maximum velocity in the flow transitions from the pile-up location to the keel. Rosso et al.^[43] investigated the oblique and asymmetric water entry problem through a series of experiments on a rigid wedge. They utilized particle image velocimetry (PIV) to analyze the influence of heel and velocity angles on the evolution of the pile-up and hydrodynamic loads. The asymmetric impact loads and hydroelastic responses of a ship model were experimentally investigated by Lin et al.^[44], presenting nonlinear springing and whipping responses in oblique wave conditions. It was also observed that the distributions of peaks of horizontal and torsional loads exhibit highly asymmetric properties. The effect of heel and pitch angles on the hydrodynamic pressure of a truncated ship bow

under asymmetrical impact was experimentally studied by Xie et al.^[45]. Their experimental study was further extended to investigate the hydroelastic slamming of a thin-welded structure under oblique impact Xie et al.^[46]. Hosseinzadeh et al.^[47] provided a detailed experimental analysis on the impact-induced loads and structural responses of a 3D non-prismatic aluminum wedge with stiffened plates. They examined the significance of hydroelasticity by analyzing the hydrodynamic pressure and strain responses in detail.

In summary, the examination of past research highlights the need for further investigation into the intricacies of oblique slamming on 3-D structures. This paper contributes to addressing this gap by presenting a series of experimental study on the dynamic response of a 3-D wedge section. The investigation specifically focuses on the hydroelastic effects arising from asymmetric slamming loads, exploring into their impact on hydrodynamic pressure and asymmetric acceleration at varying drop heights. A key feature of the study is the use of a wedge composed of both stiffened and unstiffened plates, allowing for a detailed assessment of how structural flexibility influences the obtained results. This research aims to enhance our understanding of the complex dynamics associated with oblique slamming on 3-D structures.

1. Experimental setup

This series of experiments focuses on a 3-D wedge section that resembles the fore body structure of moderate to high-speed crafts. The experimental setup was meticulously devised and constructed at the Marine Technology Competence Centre (MARTE) of Tallinn University of Technology, featuring a towing tank with 60 m length, 5 m breadth and 3 m water depth. The wedge design aimed to achieve a balance between practicality for laboratory scale production and ensuring sufficient stiffness disparity between the port and starboard sides of the specimen. To attain this, the bottom of the wedge section was intentionally designed with varying deadrise angles ranging from 20° to 30°. The starboard of the wedge was constructed using a 4 mm thick extruded aluminum panel with a single T-shaped longitudinal and transverse stiffener, while the port side was left unstiffened. The designed wedge has a mass of 55 kg including all

Table 1 Particulars of the aluminum wedge

($L \times B \times H$)/m	Density, ρ_{Al} (kg/m ³)	Modulus of elasticity, E /GPa	Shear modulus, G /GPa	Poisson ratio, ν	Tensile strength /MPa	Longitudinal and transverse stiffener/mm	Keel flat bar/mm	Thickness of the endplates/ mm	Thickness of the side plates/mm
1.5×0.94×0.45	2 700	68	26	0.33	300	T54×3+T35×4	60×5	10	4

sensors and attachments. The characteristics of the wedge, including its material properties and dimensions, are summarized in Table 1. This set of drop tests provides a detailed exploration of the hydrodynamic behavior of the asymmetric water entry, offering valuable insights into the effects of varying deadrise angles and structural stiffness on the dynamic response.

1.1 Test tower

The drop test setup was designed to facilitate experiments across a wide range of heel angles. In Fig. 1, a schematic view of the test tower is presented. The test rig is a modification of the test frame introduced by Hosseinzadeh et al.^[47], specifically adapted for asymmetric drop tests. An angle adjustment mechanism was incorporated on fore and aft of the wedge to configure the desired heel angle. This test tower lets us set the desired drop height and heel angle easily for conducting asymmetric wedge slamming experiments. As depicted in Figure 1, the test can be conducted up to 1.5 m, however, in this series of experiments, we limited the tests to a maximum of 50 cm. The construction and installation of the test tower within a segment of the towing tank were executed with a focus on rigidity to minimize its impact on experimental results. A manually operated winch, aligned with the center axis of the test tower, was employed for lifting the wedge to the designated drop height.

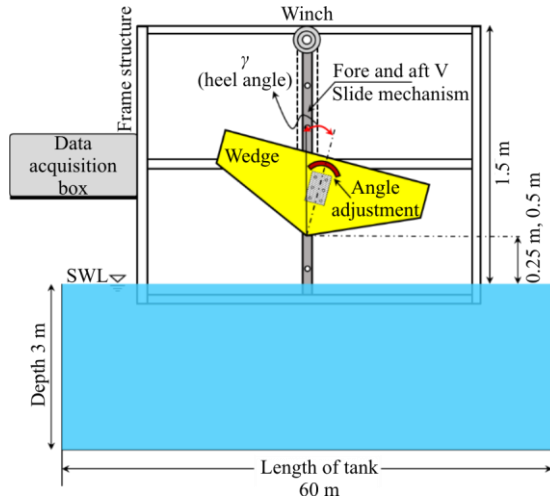


Fig. 1 (Color online) Schematic view of test set up

1.2 Sensor arrangement and DAQ system

During the experiments, a set of sensors was employed to capture essential data. The asymmetric acceleration was measured at three different locations on the wedge section to analyze the influence of the deadrise angle. Figure 2 illustrates the placement of three Dytran 3176B piezoelectric accelerometers

installed to the top of the keel. These accelerometers are positioned at the fore, middle, and aft of the section, denoted by A_f ($\beta = 30^\circ$), A_m ($\beta = 25^\circ$) and A_a ($\beta = 20^\circ$). This strategic arrangement allows for a comprehensive assessment of asymmetric acceleration across different locations of the wedge section, providing a better understanding of the effect of varying deadrise angles on the structure.

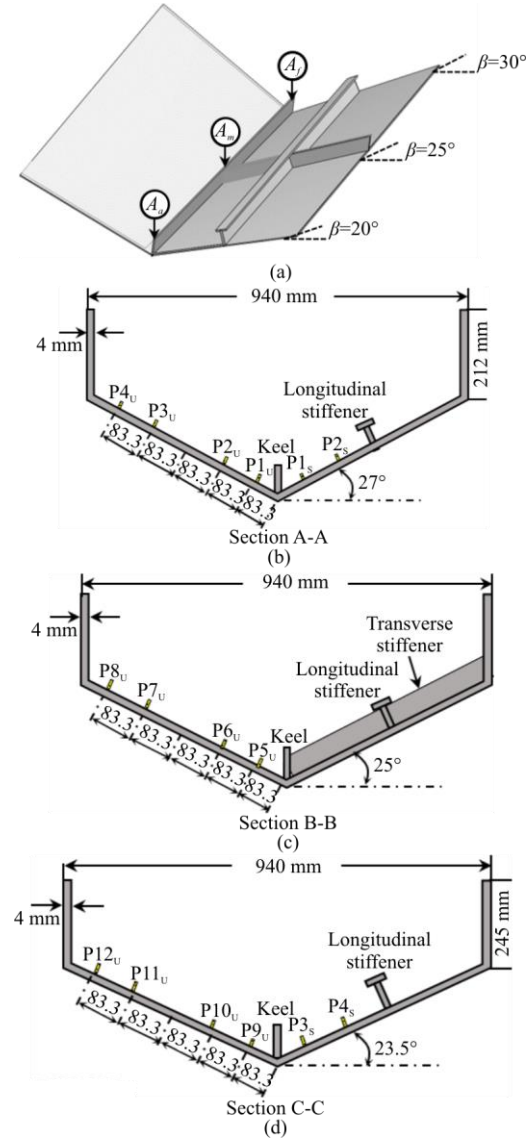


Fig. 2 A view of sensor arrangement at different cross sections of the model at upright position ($\gamma = 0$)

In addition to the accelerometers, the experiment utilized sixteen piezoelectric pressure transducers (PCB-CA102B18) to measure the impact pressure with high precision. These miniature dynamic pressure sensors provide the advantage of recording data accurately within the brief duration of slamming impact. These pressure sensors were strategically

placed to allow measurement at various locations on the wedge. As shown in Figure 2, the pressure sensors are symmetrically positioned on both sides of the wedge to facilitate a comparison between the results obtained from stiffened and unstiffened bottom plates.

To capture pressure along the length of the wedge and assess three-dimensional effects, the sensors were distributed along three characteristic lines from keel to chine, with a longitudinal distance of 166.6 mm between them. Figure 2 reveals an absence of pressure sensors on the stiffened side at a 25° deadrise angle due to the transverse stiffener preventing sensor mounting. Consequently, there are twelve pressure sensors on the unstiffened side and four sensors on the stiffened side of the wedge. These pressure sensors, denoted by P, are labelled differently on the starboard and port sides of the wedge. For example, P1_S indicates the first pressure sensor on the stiffened bottom, while P1_U refers to the first pressure sensor on the unstiffened bottom.

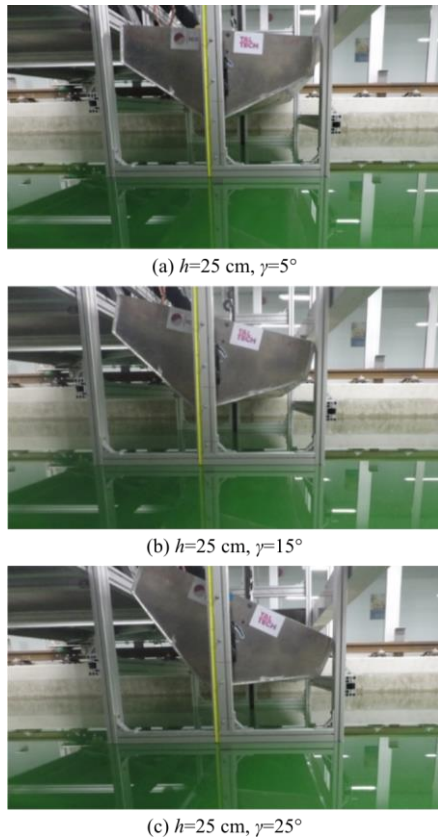


Fig. 3 (Color online) Initial position of the wedge at different heel angles

All measured data were collected using a data acquisition system installed on the towing tank carriage. Given the rapid nature of slamming impact, occurring within milliseconds to microseconds at high impact velocities, a high sampling rate data collection

device was imperative. To capture hydrodynamic pressure data, a highly dynamic universal amplifier with a sampling rate of 100 kHz was employed. The accelerometers were configured with a sampling rate of 40 kHz. The HBM CX22B-W data recorder module, connected to the HBM proprietary CatmanEASY AP software, facilitated the collection of sensor data. This approach ensured accurate and high-frequency data acquisition, crucial for analyzing the dynamic and fast-paced events associated with slamming impacts.

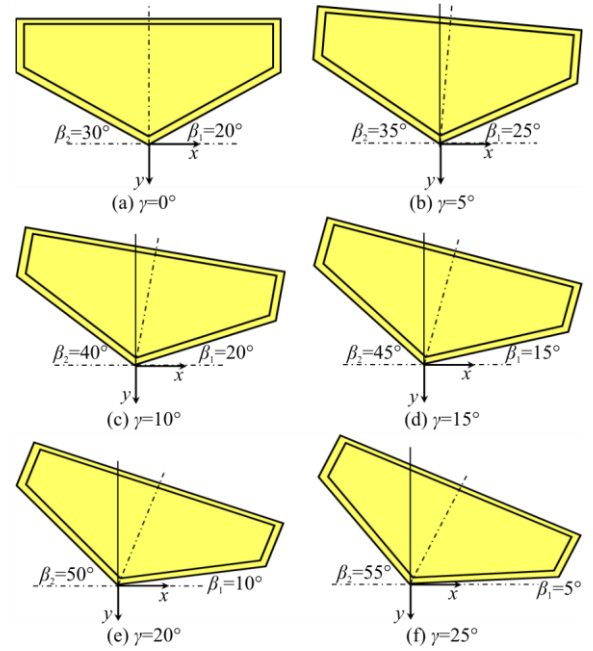


Fig. 4 (Color online) Schematic view of the wedge section of the side with 30° deadrise angle at various heel angles (Note: left side—stiffened bottom and right side—unstiffened bottom)

1.3 Experiment procedure

This series of experiments included two different drop heights, 25 cm, 50 cm, across various heel angles, as detailed in Table 2. At the outset of each test, the wedge was positioned at the desired heel angle and subsequently raised to the designated drop height. The precise location of the drop height was determined using a laser level, a movable stopper, and a measurement tape attached to the test tower.

During the experiment, determining the heel angle is crucial due to the variable deadrise of the wedge section along its length. Consequently, the angle of attack differs along each characteristic line where a set of pressure sensors is installed. Figure 3 illustrates an example of a section with a 30° deadrise angle. Notably, at different heel angles, the right and left sides of the wedge exhibit varying deadrise angles denoted by β_1 and β_2 for the unstiffened and stiffened bottom. The initial positions of the wedge at

different heel angles are depicted in Fig. 4, providing a visual representation of the experimental setup.

Table 2 Test matrix for different conditions

Run No.	Drop height	Heel angle/ $^{\circ}$
1	25 cm	5
2	25 cm	10
3	25 cm	15
4	25 cm	20
5	25 cm	25
6	50 cm	5
7	50 cm	10
8	50 cm	15
9	50 cm	20
10	50 cm	25

In this section, a comprehensive overview of the experimental setup has been provided, detailing the equipment used, including sensors and the data acquisition system. This detailed setup ensures the collection of precise and high-frequency data during slamming events, enabling a thorough evaluation of hydrodynamic behaviour. In the next section, the acquired results will be presented and thoroughly discussed to draw meaningful insights from the experimental findings.

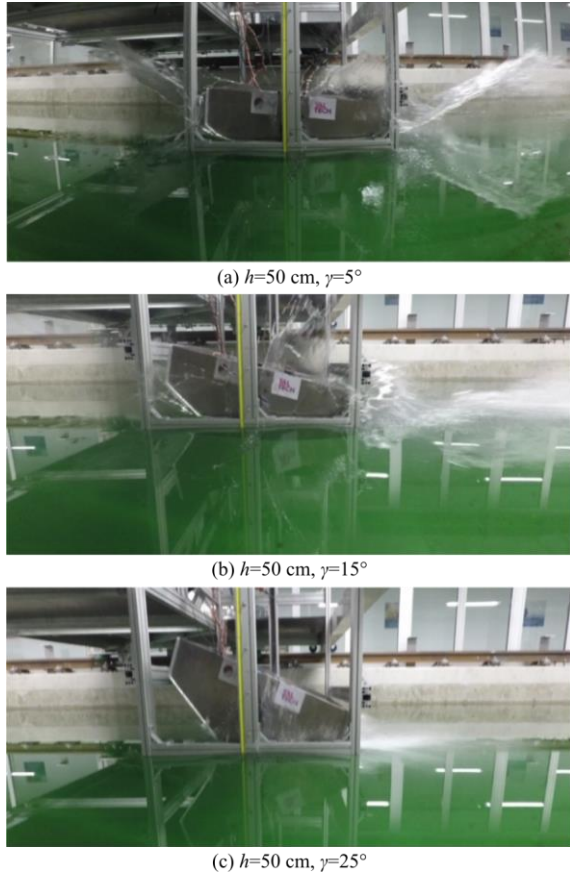


Fig. 5 (Color online) Maximum penetration of the wedge section of the test case with 50 cm drop height and different heel angles

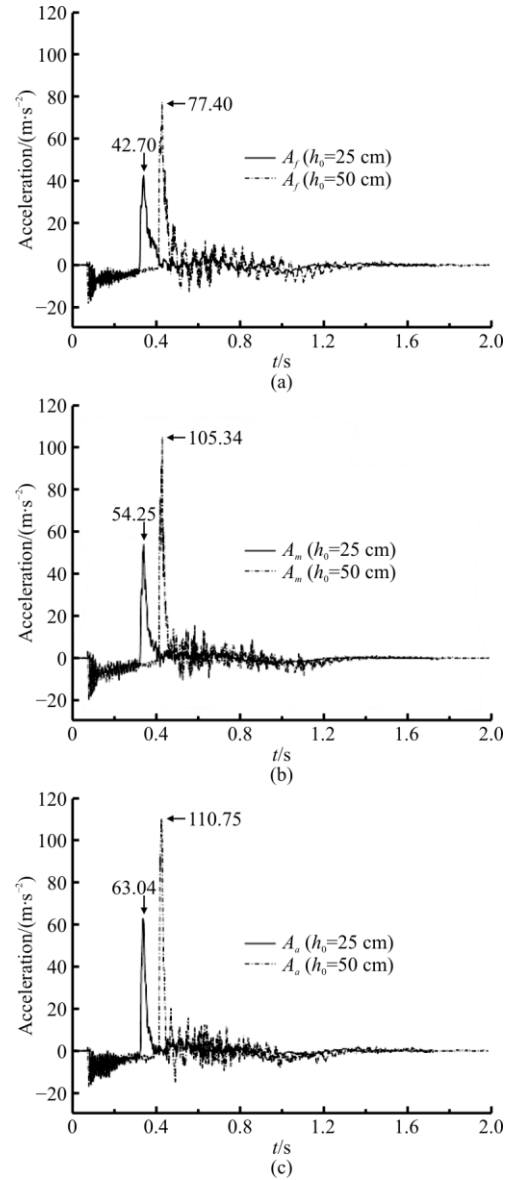


Fig. 6 Comparison of the asymmetric acceleration between two drop heights, each with a heel angle of 15° , and across three different locations on the wedge

2. Results and discussions

2.1 Impact acceleration

This section presents a comprehensive analysis of experimental results obtained under various conditions. Figure 5 visually depicts the water impact of the test case at an initial drop height of 50 cm with different heel angles. Notably, an observable change in the water entry impact becomes apparent with increasing heel angles, influencing the dynamics of the structure. To assess drop test repeatability and gauge experimental data uncertainty, each run was repeated at least twice. The presented data represents

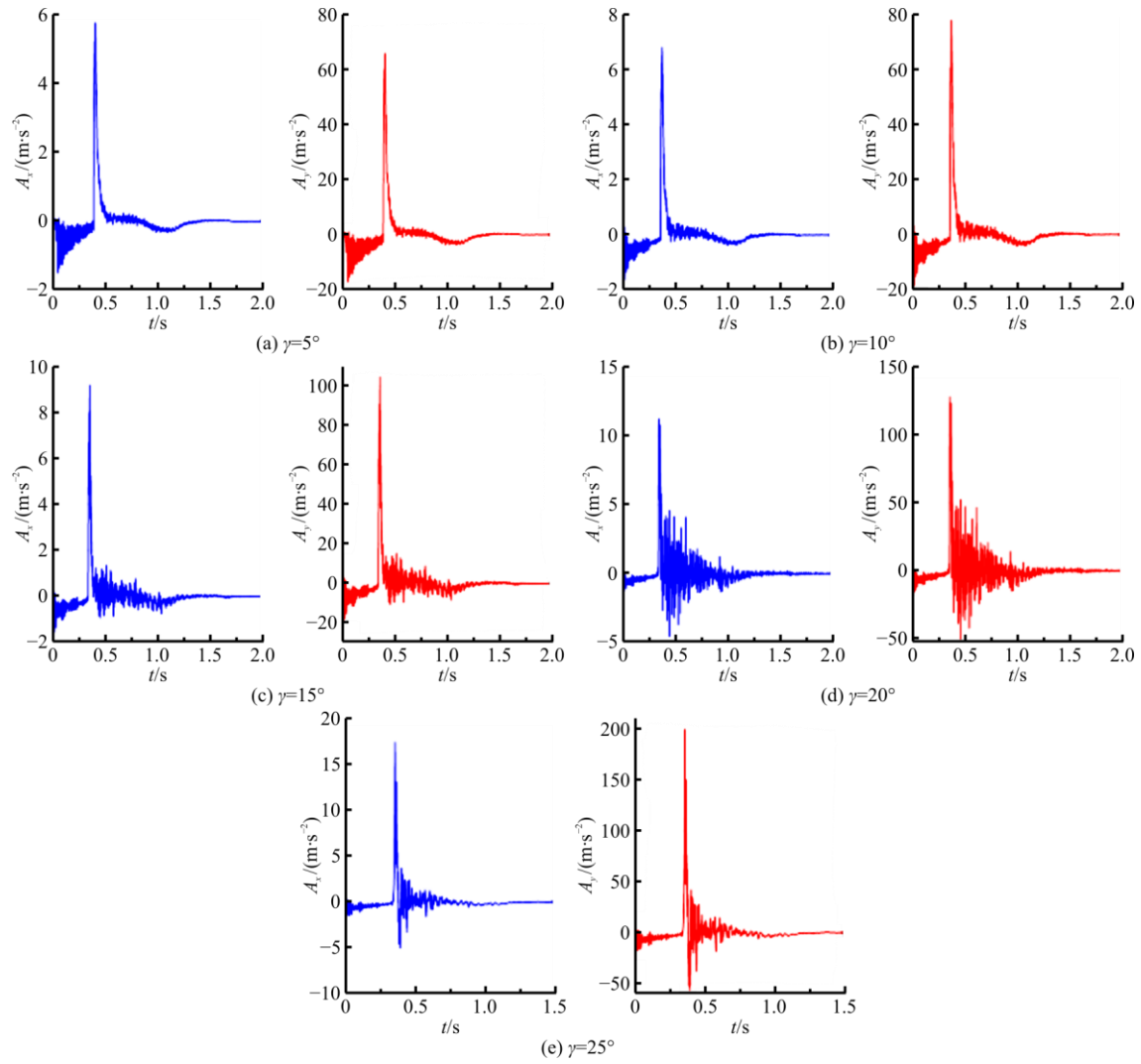


Fig. 7 (Color online) Effect of heel angle on the horizontal (A_x) and vertical (A_y) acceleration at the middle of the wedge for the case of 50 cm impact

the mean value of these repeated runs. Furthermore, the relative standard deviation of the recorded data from all sensors has been discussed in detail in Hosseinzadeh et al.^[47]. The comparison of experimental outcomes under these various conditions serves as a basis for in-depth insights into the asymmetric slamming phenomenon.

As mentioned in the previous section, the wedge section was made up with varying deadrise angles. To evaluate the impact of deadrise angle on asymmetric acceleration, three accelerometers were installed on the top of the keel at different longitudinal positions—fore, middle, and aft of the wedge. Analyzing the obtained results revealed that the highest peak acceleration occurs at the accelerometer situated on the section with a deadrise angle of 20° (A_y). In Figure 6, a comparison is presented, showcasing the asymmetric acceleration of the wedge under two different initial

drop heights at a 15° heel angle. The interaction of the wedge with the calm water surface induces a quick acceleration peak that reached within a short period of time. As expected, the velocity of impact has a notable influence on the maximum acceleration, with an increase in drop height leading to a higher peak value of acceleration. The maximum values of asymmetric acceleration at a 25 cm drop height are 42.70 m/s^2 , 54.24 m/s^2 and 63.04 m/s^2 , while at a 50 cm drop height, they are 77.40 m/s^2 , 106.34 m/s^2 and 110.74 m/s^2 for the sensors located at the fore, middle, and aft of the wedge section.

To explore the impact of heel angle on acceleration, both horizontal (A_x) and vertical (A_y) accelerations were extracted from the recorded data. Figure 7 illustrates a comparison of the horizontal and vertical accelerations measured from the sensor located at the middle of the wedge at various heel

angles for the case with a 50 cm initial drop height. It is demonstrated that an increasing heel angle results in a higher maximum acceleration, particularly notable in the vertical acceleration.

Moreover, examining the time history of the acceleration results reveals that the dynamic response of the structure becomes more pronounced in cases with higher heel angles. Notably, a distinct negative acceleration value is observed at heel angles of 20° and 25° (Figure 7). This implies that the heel angle can impact not only the dynamics of the slamming phenomenon but may also induce whipping and springing effects in the structure. The analysis of acceleration data, both horizontal and vertical, provided valuable insights into the dynamic behaviour of the wedge structure during water entry. These findings contribute to a deeper understanding of the complex dynamics involved in oblique water entry, shedding light on potential whipping and springing effects induced by varying heel angles.

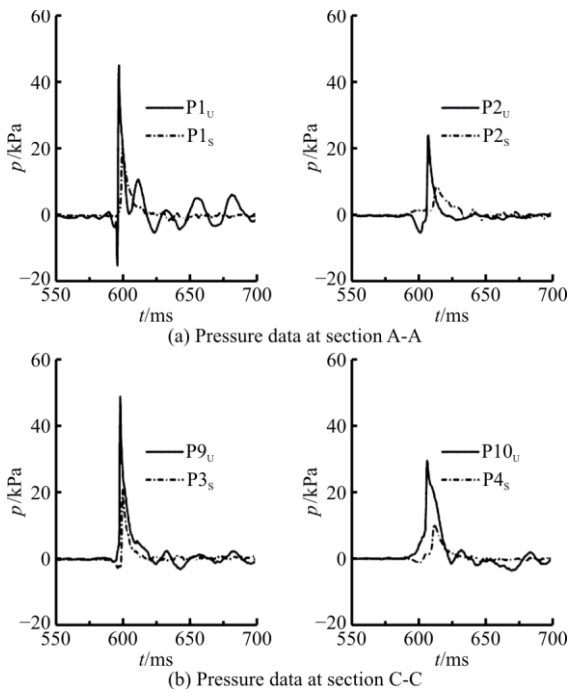


Fig. 8 Comparison of the stiffened and unstiffened measured pressure data at $\gamma = 15^\circ$ and 50 cm drop height

2.2 Hydrodynamic pressure

In addition to the analysis of acceleration, the measurement of slamming pressures was conducted at various locations along the wedge section. To investigate the influence of structural rigidity on hydrodynamic pressure, the wedge was intentionally designed with two different bottom plates, referred to as the stiffened (S) and unstiffened (U) sides. Figure 3 illustrates the placement of the pressure transducers at different sections of the wedge, where the deadrise

angle varies. This design enables the investigation of how differing structural configurations impact hydrodynamic pressures, providing valuable insights into the relationship between structural properties and slamming forces during asymmetric water entry.

Figure 8 provides a comparison of the time histories of pressure results between the stiffened and unstiffened bottom plates at two distinct. This figure presents the results of the test case with initial drop height is 50 cm and a 15° heel angle. While higher pressure on the stiffened bottom is expected, the varying impact angles on each side of the wedge, resulting from the applied heel angle, introduce another scenario. In this series of drop tests, the unstiffened bottom is considered as the side that interacts with the water surface first.

In addition to the maximum pressure values, the dynamic loads exhibit rapid changes, propagating through hydrodynamic pressure and inducing elastic vibrations. Figure 8 clearly illustrates that elastic vibrations are more pronounced on the unstiffened plate compared to the stiffened plate. The results from sensors located on the stiffened bottom show no clear evidence of elastic vibrations.

The comparison of maximum pressure values on various locations of the unstiffened side of the wedge is depicted in Figure 9. The x -axis illustrates the considered heel angles throughout the experiment. An analysis of the impact of different heel angles on the pressure peak value is conducted. The results indicate that increasing the heel angle leads to higher impact pressure. However, in some instances, the pressure value decreases, especially for heel angles exceeding 20°. This phenomenon is attributed to the non-prismatic nature of the designed wedge structure. For instance, when the heel angle equals 25°, the deadrise angle between the pressure sensors and water surface becomes zero, and that particular section behaves like a flat plate. This effect is more pronounced for the sensors positioned on the wedge section with a lower deadrise angle (P9-P12).

Figure 9 additionally provides an analysis of the maximum pressure values for two drop heights. The vertical axis of the plots is normalized to facilitate result comparison. It is evident that higher impact velocity leads to increased peak pressure. However, for higher heel angles, the scenario may differ because the angle between the wedge and water surface approaches zero. This introduces other phenomena that needs further investigation, such as the role of aeration and cavitation on the impact induced loads of the elastic structures.

In addition to the pressure results on the unstiffened side of the wedge, Figure 10 presents the maximum pressure measured on the stiffened side. The pressure peak values are compared at different

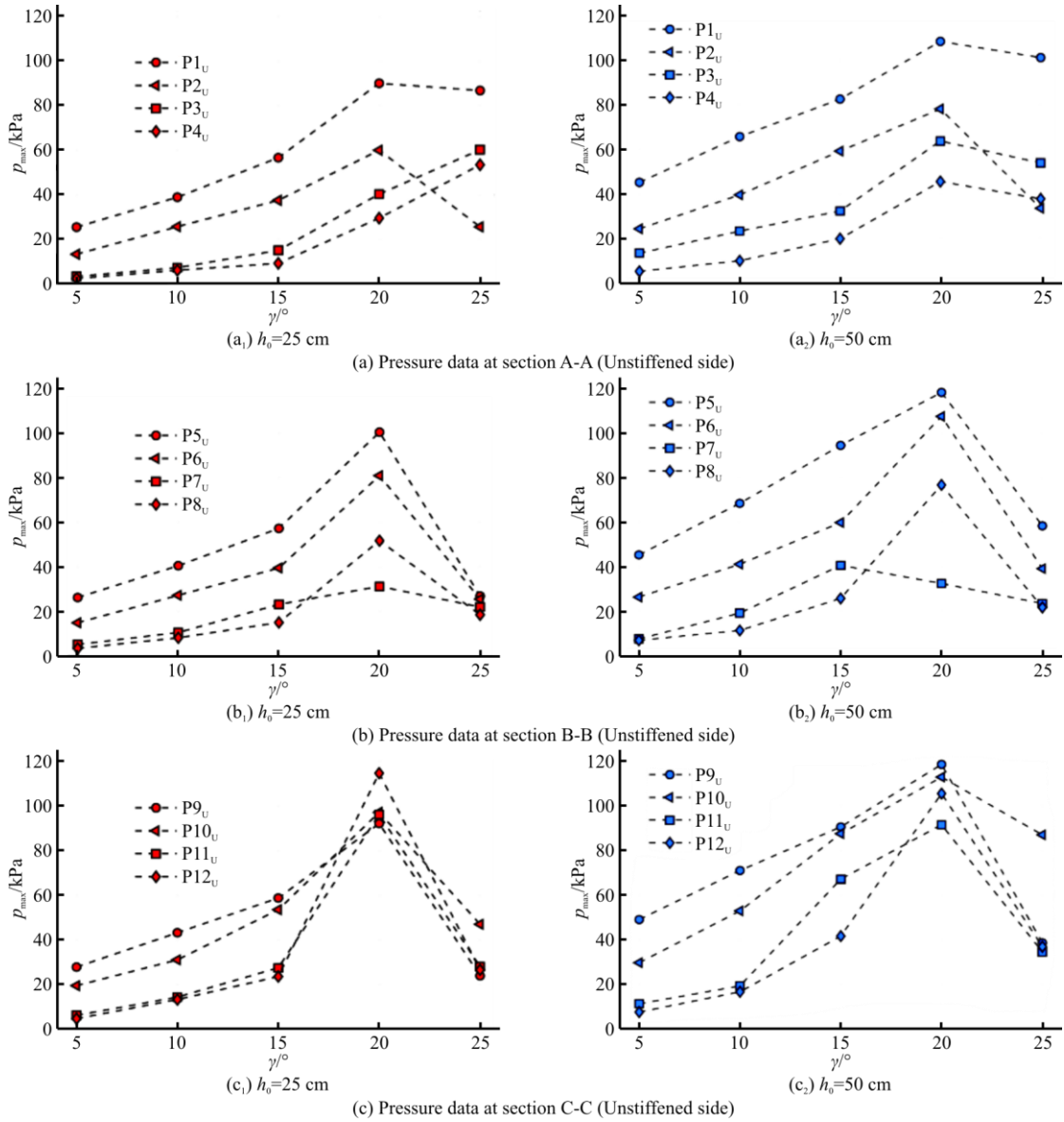


Fig. 9 (Color online) The effect of heel angle on the maximum pressure values of the unstiffened bottom at three different sections of the wedge

heel angles and two drop heights. As expected, the pressure results at section C-C are higher than those of section A-A that is the section with higher deadrise angle. This means that the hydrodynamic pressures are very sensitive to the change of deadrise angle.

The impact pressure on both the unstiffened and stiffened sides of the wedge has been compared, examining both the time history and maximum values. These results offer valuable insights into the influence of the heel angle on hydrodynamic pressures during the asymmetric water entry impact.

3. Conclusions

Impact-induced loads and dynamic responses

resulting from slamming events are critical aspects to account for in the design of different marine floating structures. Obtaining high-quality experimental data on slamming loads and responses is crucial. This data not only aids in developing a more profound comprehension of hydroelasticity but also contributes to refining current theories and validating numerical simulations. It plays a vital role in ensuring the effectiveness and reliability of calculating slamming loads for various marine structures.

This paper examined and compared the impact-induced loads acting on a 3-D V-shaped elastic structure. Experimental analysis of asymmetric slamming was conducted using a non-prismatic aluminum wedge section. One side of the wedge

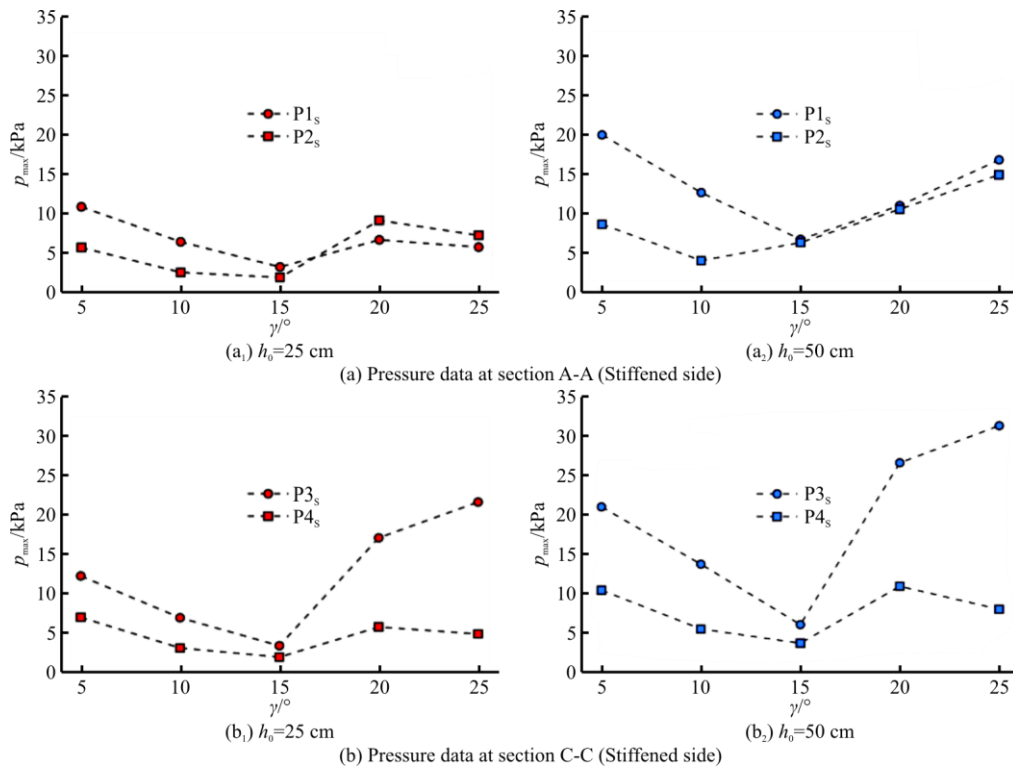


Fig. 10 (Color online) The effect of heel angle on the maximum pressure values of the stiffened bottom at two different sections of the wedge

featured a stiffened panel, while the other side remained unstiffened to explore the structural rigidity. The influence of impact velocity on acceleration was investigated, along with the impact of heel angle on both vertical and horizontal acceleration. The findings revealed that an increase in heel angle led to higher impact acceleration, particularly noticeable in the vertical direction. Slamming pressure was assessed at varying heel angles on both stiffened and unstiffened sides. Higher pressures were observed on the side facing the oblique fluid, resulting in more pronounced dynamic responses. The time history of pressure results highlighted the impact of elastic vibrations, particularly evident on the unstiffened bottom plate.

The study results presented in this study have deepened our understanding of the asymmetric slamming effects on aluminum structures. Future research could focus on exploring the impact of heel angle on structural responses, as well as developing analytical models for elastic structures and potentially validating numerical models.

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Compliance with ethical standards

Conflict of interest: The authors declare that they have no conflict of interest. All authors declare that there are no other competing interests.

Ethical approval: This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent: Not application.

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